



Review

Advances in the critical care management for patients with hematological malignancies

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ARTICLE INFO

Keywords:

Mechanical ventilation
Leukemia
Neutropenia
Transplantation
Aspergillosis
Pneumocystis

ABSTRACT

Hematological malignancies (HMs) are increasingly associated with life-threatening complications requiring intensive care unit (ICU) management. Recent advancements in therapies, diagnostics, and critical care protocols have improved outcomes for these patients, yet significant challenges persist. This manuscript explores the evolving landscape of critical care in hematology, emphasizing the unique complications, management strategies, and future directions in the field.

Patients with HMs are particularly vulnerable to infections, sepsis, organ dysfunction, and treatment-related toxicities such as cytokine release syndrome (CRS), immune effector cell-associated neurotoxicity syndrome (ICANS), and coagulopathies. Innovations in the management of acute respiratory failure, septic shock, and invasive fungal infections have contributed to better survival rates, yet outcomes remain suboptimal for certain high-risk groups. Furthermore, new therapies, including CAR-T cells, bispecific antibodies, and immune

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<https://doi.org/10.1016/j.blre.2025.101306>

checkpoint inhibitors, present both opportunities and challenges in the ICU setting due to their potential toxicities.

Emerging trends emphasize the importance of early ICU admission, multidisciplinary collaboration, and precision medicine in improving patient care. The integration of biomarker-driven strategies, advanced diagnostics, and artificial intelligence holds promise for optimizing therapeutic interventions and enhancing antimicrobial stewardship. Additionally, patient-centered approaches, including time-limited trials and goal-oriented discussions, aim to balance aggressive care with quality-of-life considerations.

This review underscores the need for continued research to address disparities in access to care, improve long-term outcomes, and develop standardized protocols for managing critically ill hematology patients. By advancing the integration of oncology and critical care, clinicians can better navigate the complexities of modern therapies and provide holistic, evidence-based care that aligns with patient values and priorities.

1. Introduction

Recent advancements in critical care have significantly improved outcomes for patients with hematological malignancies (HMs). These malignancies, including leukemias, lymphomas, and myelomas, are frequently associated with life-threatening complications requiring intensive care unit (ICU) admission. Improvements in therapies, diagnostics, and supportive care have enabled better survival rates and functional outcomes. However, challenges remain, particularly in addressing immunosuppression, infections, and treatment toxicities. This paper focuses on these developments and highlights strategies that enhance care, reduce mortality, and improve quality of life.

1.1. Patients with hematological malignancies often need an ICU bed

Hematological malignancies account for 7% of new cancer diagnoses globally, with an increasing incidence rate over the past 40 years [1]. Hematologic malignancies encompass a diverse group of conditions with distinct clinical characteristics, complications and prognosis. Broadly, these patients can become critically ill as a consequence of the underlying disease or the effects of treatments such as chemotherapy, radiation, hematopoietic stem cell transplant, or immunotherapy (Table 1).

Intense immunosuppression is a common feature for patients with hematologic malignancies which could be attributable to the disease or treatment received. There is important variability in the immune-defect associated with each disease subtype and corresponding treatments. Type, depth and duration of immunosuppression is critical to understand what types of infectious critical illness the patient may be susceptible to and to inform empiric treatment at the onset of acute/critical illness. For instance, neutropenia is frequent early following chemotherapy and in HSCT recipients before engraftment. Patients with multiple myeloma and chronic lymphocytic leukemia face immune deficits

primarily in B lymphocytes and humoral immunity, whereas T lymphocytes are more frequently affected in T-cell lymphomas and Hodgkin disease [2]. HSCT recipients present the highest complexity in this regard, where the predominant type of immune deficiency varies significantly from the time of transplant to months following engraftment. Finally, variability may also exist surrounding the duration of immunosuppression as some treatment regimen lasts months (AML) whereas others may require ongoing treatment for a few years (ALL).

Beyond the immunosuppression, patients with hematologic malignancies could present with diverse pro-inflammatory processes leading to critical illness (Table 2). These include leukostasis-induced inflammation (more frequent in AML), cytokine storm (following CAR T-cell therapy), differentiation syndrome (in patients with acute promyelocytic leukemia), peri-engraftment syndrome (HSCT recipients), tumor lysis syndrome (non-Hodgkin lymphomas, acute leukemias) and immune reconstitution (AML during cell count recovery). Furthermore, activation of the coagulation cascade can lead to both thrombotic and bleeding events. In patients with APML, coagulation activation can lead to disseminated intravascular coagulation presenting with thrombotic and bleeding events which are a common reason for ICU admission in these patients [3].

Sepsis and acute respiratory failure remain the most frequent reasons for ICU admission amongst the majority of patients with hematologic malignancies [4]. However, the relative frequency may vary depending on the underlying disease. Acute kidney injury is the most frequent reason for ICU admission among patients with multiple myeloma [5].

Table 1
Cumulative incidence and time to ICU admission in patients with hematological malignancies in the year following the diagnosis. From Ferreyro et al. Intensive Care Med 2021; 47(10): 1104-1114.

Disease group	Cumulative incidence	Median time to ICU (days) (interquartile ranges)
Acute myeloid leukemia	22.5 %	27 (3 – 101)
Aggressive non-Hodgkin lymphoma	17.7 %	28 (5-110)
Acute lymphoblastic leukemia	17.5 %	24 (6 – 128)
Myelodysplastic syndrome	16.0 %	59 (5 – 170)
Multiple Myeloma	15.0 %	45 (5 - 156)
Myeloproliferative neoplasm	12.6 %	22 (0 – 137)
Hodgkin lymphoma	9.7 %	63 (8 – 163)
Chronic lymphocytic leukemia	8.0 %	35 (1 – 175)
Indolent non-Hodgkin lymphoma	7.3 %	71 (6 – 186)

Table 2
Reasons for ICU admission, life supporting interventions and main diagnoses in patients with hematological malignancies.

Reasons for ICU admission	Proportions
Acute respiratory failure	62.5%
Shock	42%
Acute kidney injury	30%
Coma	22%
Chemotherapy in high-risk patients	7%
Life supporting interventions	
Vasoactive drugs	51%
Invasive mechanical ventilation	48%
Noninvasive mechanical ventilation	31%
Renal replacement therapy	26%
Chemotherapy	24%
Main diagnosis	
Pneumonia	36%
Septic shock	26%
Acute kidney injury	20%
Cardiac events	14%
Organ infiltration by the malignancy	13%
Typhilitic and colitis	10%
Tumor lysis syndrome	9.5%
Severe bleeding	5%
Severe metabolic disturbances	4%
Chemotherapy-induced toxicity	4%
Catheter-related Infection	3%
Coma	3%

From Azoulay et al. J Clin Oncol. 2013 Aug 1;31(22):2810-8.

Patients with lymphoma may be admitted due to airway obstruction or superior vena cava syndrome related to bulky mediastinal masses. Major bleeding is more frequent in patients with prolonged thrombocytopenia such as those with acute leukemia, HSCT recipients and myelodysplastic syndromes [6].

2. Management of hematological malignancies in the ICU

The management of critically ill hematology patients balances aggressive interventions with realistic care goals. Recent evidence highlights better outcomes with early ICU admission, multidisciplinary teams, and time-limited trials to assess treatment responses. Comprehensive care plans should integrate patient values, emphasizing palliation when recovery is unlikely while addressing symptom's control for everyone.

2.1. Patients with AML

In the past decade, advancements in AML treatment have broadened therapeutic options for patients across all ages and genetic profiles [7]. This shift has changed admission policies, particularly for elderly patients once considered palliative, who now might benefit from new therapies [8]. While intensive chemotherapy with cytarabine and anthracycline remains the standard for most younger AML patients, frontline treatment options have significantly expanded. Personalized treatment now considers patient fitness, comorbidities, and disease characteristics, such as cytogenetic and molecular profiles, including *FLT3*, *NPM1*, *IDH1/2*, and *TP53* mutations. In cases of hyperleukocytosis at diagnosis, with organ failures often requiring ICU admission, cytoreduction with hydroxyurea before induction chemotherapy is preferable to lower cell counts and reduce short-term mortality [9]. Recent data suggest that adding dexamethasone during induction chemotherapy may reduce the need for mechanical ventilation and short-term mortality [10]. Studies to explore the early use of targeted therapies (such as *FLT3* inhibitors) to rapidly manage hyperleukocytosis in ICU are warranted [11].

2.2. Patients with lymphoma

Malignant lymphomas represent a heterogeneous group of lymphoproliferative diseases, leading to ICU admission in approximately 18% of cases (Table 1) [12]. Patients with T-cell lymphoma generally face poor long-term outcomes, whereas those with diffuse large B-cell lymphoma tend to have a more favorable prognosis [13,14].

At diagnosis, compression of the heart, lungs, large vessels, nervous system, or other organs by lymphoma masses can result in life-threatening complications. Rapid diagnosis (via CT scan or biopsy) and prompt tumor reduction through chemotherapy, alongside bridging organ support, are critical. ICU admissions during the course of the disease are often therapy-associated: Tumor lysis syndrome (TLS) may occur in up to one-third of high-risk patients, particularly in B-cell and Burkitt's lymphoma [4]. The release of potassium, phosphate, and nucleic acids can lead to arrhythmias, vomiting, neurological symptoms, and acute renal failure [15,16]. Early ICU admission may improve outcomes.

Infections are common, either during chemotherapy-induced neutropenia or as a result of immunosuppression caused by the underlying disease. Sepsis frequently prompts ICU admission, and fungal or pneumocystis infections are particularly prevalent in T-cell lymphomas [13,17,18].

Additionally, lymphomas may trigger hemophagocytic lymphohistiocytosis (HLH), autoimmune hemolytic anemia, or thrombocytopenia. Complications related to immunomodulating therapies (such as CAR-T cells) include cytokine release syndrome (CRS) and immune-effector cell-associated neurotoxicity syndrome (ICANS), often requiring ICU care and specific interventions [19,20]. Given the highly specialized and

complex management of malignant lymphomas, close collaboration between intensivists and hematologists is essential.

2.3. Patients with myeloma

Approximately one out of five patients with newly diagnosed HMs presents with multiple myeloma (MM) [21]. An armada of therapeutic options including immunomodulators, chemotherapy, autologous stem cell transplantation, monoclonal and bispecific antibodies (bsAB), proteasome-, BCL-2- and XPO1-inhibitors, and CAR-T cell therapy are now associated with the prospect of long-term survival. Survival and functional outcomes after ICU are favorable and potentially even better than in other HMs [22,23].

Challenges in ICU care of patients with MM encompass the management of acute renal failure [24,25], infectious complications, part of which are associated by immunoglobulin deficiency or side-effects of therapies, and thromboembolic events. In patients with MM, the increasing number of therapeutic lines and the subsequent aging of patients leads to a greater prevalence of non-hematological comorbidities and related issues compared to other hematologic disorders. Depending on the individual risk, administration of immunoglobulins, specific anti-infective prophylaxis [26], or anticoagulation (particularly in case of immunomodulator therapy) may be indicated. Eventually, common side effects of bsABs and CAR-T cell therapy such as CRS and ICANS, as well as other, in part late neurotoxicity have recently become relevant to intensive care.

3. Emergencies in hematology

3.1. Complicated neutropenia

Case Vignette 1

A 16-year-old patient was admitted to the ICU with hyperleukocytic AML4 leukemia. Following induction therapy with anthracycline analogues and cytosine arabinoside, the patient developed profound and prolonged neutropenia. The ICU stay was complicated by multiple bacterial infections, including severe typhilitis with *E. coli* bacteremia, which was managed conservatively with antibiotics. The patient required fluid resuscitation and vasopressor support for three days. Subsequent acute respiratory failure occurred, due to invasive pulmonary aspergillosis and mucormycosis. This was treated with Isavuconazole, and high-flow nasal oxygen was administered for five days.

The patient was discharged alive three months after ICU admission. Complete remission was achieved and remains ongoing nine months post-admission.

This case illustrates risks of bacterial and fungal infections in patients with deep and prolonged neutropenia. Management emphasizes prompt antibiotics and antifungals. The use of vasopressors and subsequently high-flow oxygen with appropriate in-ICU monitoring are emphasized.

1. Acute respiratory failure

Prognosis for patients admitted to the ICU with acute respiratory failure (ARF) has improved in recent years for several reasons. Although results from retrospective studies remain to be proven, early admission before severe organ failure appears to be associated with better outcomes [27]. Advances in non-invasive diagnostic strategies have reduced the need for bronchoalveolar lavage (BAL), which was previously linked to high mortality rates [28,29]. The detection of upper airway viruses has also helped stratify the severity of ARF, enabling more tailored management [30]. Additionally, non-invasive oxygenation strategies have evolved, shifting from an emphasis on avoiding intubation for patients responding to high flow oxygen, to prioritizing earlier intubation for severe cases. This shift has contributed to a decline in mortality rates among intubated patients in recent years [31].

2. Pulmonary leukemic infiltrates in patients with AML

Leukostasis, leukemic pulmonary infiltration, and acute lysis pneumopathy (ALP) represent the three main pulmonary manifestations of leukemic infiltrates. Depending on the kinetics of patient deterioration, distinguishing between these complications can be particularly challenging in real-life clinical settings. Hyperleukocytosis, defined as a white blood cell (WBC) count exceeding $50 \times 10^9/L$ [9], is associated with pulmonary involvement in approximately 40% of cases, a condition termed leukostasis [32,33]. The underlying mechanisms include elevated leukocrit leading to high blood viscosity, interactions between leukemic cells and the pulmonary endothelium, tissue damage mediated by metalloproteinases, and abnormal oxygen consumption at the microvascular level [2]. Hydroxyurea is widely recommended by international guidelines as a first-line therapy [9]. Although leukapheresis can rapidly reduce the WBC count, it does not resolve pre-existing vascular cellular plugs or inhibit further cellular proliferation. Meta-analyses suggest that leukapheresis does not reduce mortality [34,35]. Corticosteroids might be added.

Leukemic pulmonary infiltration (LPI) describes the extravascular presence of leukemic cells within the lung parenchyma, which may occur independently of hyperleukocytosis [36,37]. Although LPI does not invariably lead to acute respiratory failure (ARF), it can do so when accompanied by other pulmonary complications. Leukemic cells preferentially infiltrate alveolar septa, peribronchiolar, perivascular, interlobular, and subpleural spaces, likely due to their affinity for lymphatic routes [38,39]. While diagnosing LPI does not alter the general management of acute myeloid leukemia (AML), it prompts clinicians to initiate induction therapy without delay.

Acute lysis pneumopathy (ALP) is characterized by ARF following a rapid decrease in blast cell count in patients with hyperleukocytosis. ALP may worsen the respiratory status of patients with leukostasis or LPI, potentially leading to diffuse alveolar damage. There is no specific curative treatment for ALP, but spontaneous improvement has been observed, often coinciding with the resolution of metabolic abnormalities associated with tumor lysis syndrome [40].

3. Acute respiratory distress syndrome

Case vignette 2

A 61-year-old female was referred to the ICU for acute respiratory failure. She had been recently diagnosed with acute myeloid leukemia (AML) and had started induction chemotherapy 15 days prior to admission. The patient required intubation shortly after arrival. A chest X-ray revealed bilateral infiltrates, and the PaO_2/FiO_2 ratio was 134 mmHg, indicating moderate acute respiratory distress syndrome (ARDS). Her body temperature was 39°C, and she was noted to be profoundly neutropenic with a neutrophil count of zero. Early diagnosis of ARDS and prompt management that include optimization of oxygenation and identification of the cause of pulmonary involvement are important to reduce mortality.

Acute respiratory distress syndrome (ARDS) represents the most severe stage of acute respiratory failure [41]. Approximately 80% of immunocompromised intubated patients meet the criteria for ARDS [42]. In patients with hematologic malignancies, ARDS exhibits specific features, including a high prevalence of fungal infections, disease-related infiltrates, and cases of undetermined etiology [42–44]. Notably, ARDS caused by fungal infections or of undetermined origin is associated with higher mortality rates, underscoring the critical importance of early diagnostic strategies and, in some cases, empirical therapeutic interventions. As with other ARDS patients, those with hematologic malignancies should receive lung-protective ventilation, including: low tidal volumes, tailored positive end-expiratory pressure (PEEP), limited plateau and driving pressures, and prone positioning when the PaO_2/FiO_2 ratio < 100–150 mmHg [42]. Finally, it is important

to recognize that mortality in ARDS patients is higher among those with hematologic malignancies compared to patients without cancer [42,44,45].

4. Septic shock

Septic shock remains a therapeutic challenge in patients with hematological malignancies due to their immune vulnerability, clinical heterogeneity, variability in clinical practices (e.g., practitioner biases and ICU/hospital policies), and rising rates of infections caused by multi-drug resistant organisms [46,47]. Although mortality rates have decreased over time [48], they remain high, particularly in patients with elevated lactate levels who meet Sepsis-3 criteria [49].

Several factors have been associated with improved survival, including higher albumin levels (but not albumin administration), the use of aminoglycosides, and administration of G-CSF. Conversely, lower survival rates have been linked to concomitant respiratory failure, higher severity of illness, and elevated lactate levels—but not neutropenia [49,50]. The need for mechanical ventilation, renal replacement therapy, or allogeneic hematopoietic stem cell transplantation has also been associated with poorer outcomes [50]. Importantly, early recognition and ICU admission are key strategies to improve outcomes [27]. Additionally, performance improvement programs, a personalized approach to hemodynamic resuscitation, and the integration of modern monitoring and diagnostic technologies contribute to better survival in this complex population [46].

5. Platelet refractoriness

Case Vignette 3

A 55-year-old man with acute myeloid leukemia (AML) and recent induction therapy was admitted to the ICU with septic shock. On admission, his platelet count was 17 G/L. Following intubation, the patient developed airway bleeding. He was treated with red blood cell (RBC) and platelet transfusions. Although the bleeding subsided, it did not completely stop. By the following morning, the patient's platelet count had dropped further to 5 G/L.

Platelet refractoriness is most often attributed to clinical factors rather than alloimmunization [51]. In cases of hypoproliferative thrombocytopenia, peripheral platelet consumption caused by severe infections frequently leads to high rates of refractoriness, affecting approximately one-third of patients [52]. This condition is associated with worse clinical outcomes.

Clinical management should prioritize addressing the underlying condition, as attempting to maintain a predefined platelet count threshold through repeated transfusions is unlikely to be effective until the infection is controlled. A careful assessment of bleeding location and severity is essential, along with consideration of local treatments, such as pharyngeal packing.

Antifibrinolytic agents may also be considered; however, recent randomized controlled trials (RCTs) on prophylactic tranexamic acid in patients with hematological malignancies have not demonstrated a reduction in bleeding [53]. Fractionated platelet transfusions is part of the strategies to consider in case of refractoriness [54].

6. Coagulopathy

Coagulopathy in critically ill hematology patients includes, but is not limited to, disseminated intravascular coagulation (DIC). Increased fibrinolysis is the hallmark of inaugural APL but can also be seen in many other types of hemopathies (AML, ALL, NHL, NPH, and others) [55]. Beyond DIC, a wide spectrum of biological anomalies (primary fibrinolysis, thrombotic microangiopathy, anti-thrombin 3 deficiency, and others) and clinical events (arterial or venous thromboembolism, spontaneous hemorrhages, and others) related to coagulopathy can be observed in hematology patients [56]. Coagulopathy can occur as a

direct consequence of active cancer or can be triggered by infections and treatments (L-asparaginase, proteasome inhibitors, tyrosine kinase inhibitors, heparin, CAR-T cells therapy, and others) [57]. Prevention and treatment of coagulopathy depends on the type, severity and presumed mechanisms of the clinical and biological manifestations [58,59].

7. Cytopenia and organ dysfunction

The prognosis of hemophagocytic syndromes (HS) in critically ill patients has not improved in recent years, despite improvements in knowledge and skills. The major challenge remains the formal identification of the diagnosis in the absence of a reliable gold standard, particularly in critically ill patients, due to numerous confounding factors. Moreover, the time window for aggressive diagnostic exploration (including biopsies, for example) is often compromised by worsening pathology (due to cytopenias and organ failure), which can lead to a "race against time." [60] Etoposide remains the gold standard in the most severe patients [61]. The administration of anti-interferon-gamma antibodies (emapalumab) has been rather disappointing in children [62]. The JAK1/JAK2 inhibitor ruxolitinib could change the prognosis of HS patients in the years to come, as it has been associated with improved HS manifestations and survival in a preclinical mouse model [63]. (3). Human data are scarce but promising, and larger studies are needed [64].

8. Hyperviscosity syndrome (HVS)

Case vignette 4: A 65-year-old man presented with a two-week history of bilateral blurry vision accompanied by epistaxis, dizziness, and somnolence. He was admitted to the ICU in a state of coma. Fundoscopy revealed retinal vein dilation, tortuosity, and retinal hemorrhages. Additional investigations showed: Mild anemia (Hb: 10 g/dL), normal brain CT scan, and a prominent monoclonal IgM-kappa band (35 g/L) identified through serum electrophoresis and immunofixation. A bone marrow biopsy was performed, confirming a diagnosis of Waldenström's macroglobulinemia.

Hyperviscosity syndrome (HVS) is a rare complication of monoclonal gammopathies, particularly those involving IgM due to its pentameric structure and thus higher molecular weight. It is most commonly associated with Waldenström's macroglobulinemia. The classic triad of HVS includes mucosal bleeding, visual disturbances, and neurological abnormalities [65]. There is no specific viscosity threshold to guide the initiation of treatment. Red blood cell (RBC) transfusions should be avoided in HVS as they may increase the risk of cerebral hemorrhage. For patients with severe neurological impairment, urgent plasma exchange with albumin is the treatment of choice. Typically, three sessions are sufficient for IgM-related HVS, as IgM is predominantly intravascular. In contrast, for IgG-mediated HVS, which involves IgG stored primarily in tissues, the third session is delayed to allow time for IgG to redistribute from the tissues to the vascular compartment. In emergencies, hyperhydration and plasma exchange are considered the standard of care. Additionally, specific treatment of the underlying malignancy is crucial for long-term management.

9. Acute graft versus host disease (aGVHD) and critical care illness

Approximately 25% of patients undergoing allogeneic stem cell transplantation (allo-HSCT) require ICU admission due to life-threatening complications, including severe infections, immune-mediated reactions, and treatment-related toxicities [12,66,67]. Recent advancements in hematology and critical care have significantly improved outcomes for these critically ill patients, with ICU survival rates now exceeding 75% [68]. Notably, a prior ICU stay is no longer associated with negative impacts on long-term survival, emphasizing the importance of effective ICU admission policies to optimize patient care.

Upon ICU admission, key mortality determinants should be

systematically assessed to align goals of care with patient preferences. In addition to organ failures and intrinsic factors—such as age, time since allo-HSCT, and comorbidities—acute graft-versus-host disease (aGVHD) remains a critical prognostic factor, especially in case of corticosteroid-resistance [69,70]. The increased vulnerability to critical illness associated with aGVHD is driven by direct tissue involvement, heightened immunosuppression, treatment-related toxicities, and malnutrition [69]. These factors collectively contribute to worse outcomes in this patient population.

10. Acute organ dysfunction in a CAR T-cell recipient

Case vignette 5: A 58-year-old male with refractory diffuse large B-cell lymphoma underwent CD19 CAR T-cell therapy following failure of R-CHOP and salvage therapies. At day 3, the patient developed a low-grade fever (38.5°C) and showed a mildly elevated CRP. At day 5, his condition worsened, presenting with a high fever (40°C), tachycardia, and hypotension requiring vasopressor support. Laboratory findings revealed a ferritin level of 12,000 ng/mL and elevated IL-6. He was diagnosed with Grade 3 cytokine release syndrome (CRS) and treated with tocilizumab, followed by transfer to the ICU. At day 6, the patient exhibited neurological deterioration with lethargy, confusion, and intermittent speech difficulty, though CT imaging showed no intracranial abnormalities. He was diagnosed with Grade 3 immune effector cell-associated neurotoxicity syndrome (ICANS) and treated with dexamethasone. Supportive care included fluid resuscitation, vasopressors, oxygen therapy, and infection prophylaxis. Due to lack of symptom improvement, anakinra was added. The patient demonstrated gradual neurologic recovery by Day 10, allowing for the tapering of steroids and anakinra. He was transferred back to the oncology ward for further care. A subsequent restaging exam revealed complete remission.

Effective management of CRS and ICANS in the ICU requires early recognition, ruling out infection, prompt initiation of targeted therapies like tocilizumab (for CRS) and corticosteroids (for ICANS), and supportive care to prevent multi-organ dysfunction. To optimize outcomes close monitoring of inflammatory markers, organ function, and neurologic status is essential, along with infection prophylaxis and multidisciplinary management. Importantly, challenges persist about the lack of reliable criteria for distinguishing CRS from sepsis, the increasing use of novel therapies such as anakinra in ICANS despite limited evidence, and the particular challenge of maintaining CAR-T cell function and expansion while managing related toxicities, which can sometimes be contradictory goals.

11. Cardiac toxicities

Case vignette 6: A 50-year-old man with diabetes, hyperlipidemia, and diffuse large B-cell lymphoma, presented with progressive dyspnea and lower extremity edema two days prior to admission. He was hypoxemic on arrival, requiring high-flow nasal cannula, and had diffuse ST-segment elevations on ECG, elevated troponins and pro-BNP, and a severely depressed ejection fraction of 20% on bedside echocardiogram. The absence of ischemic findings on emergent cardiac catheterization and a negative infectious workup strongly suggested Immune Checkpoint Inhibitors-related myocarditis, supported by his recent pembrolizumab exposure. High-dose corticosteroids were initiated promptly, the cornerstone of treatment for this condition. Despite this, the patient rapidly deteriorated, developing cardiogenic shock, atrial fibrillation, renal and respiratory failure, and refractory ventricular tachycardia. Intravenous immunoglobulin (IVIG) was administered as a second-line therapy, but the progression of multiorgan dysfunction proved irreversible, and the patient succumbed to cardiac arrest despite resuscitation efforts.

This case highlights the devastating potential of immune checkpoint inhibitor (ICI)-associated myocarditis, a rare but life-threatening complication of pembrolizumab, a PD-1 inhibitor. Diagnosing this

condition is challenging, as it requires ruling out other common etiologies such as acute coronary syndromes and infectious myocarditis. While cardiac biopsy remains the diagnostic gold standard, treatment should not be delayed in cases with high clinical suspicion. The combination of corticosteroids and additional immunomodulatory agents, such as IVIG or newer therapies like mycophenolate mofetil or infliximab, may improve outcomes in steroid-refractory cases. However, prognosis remains poor, particularly in patients presenting with fulminant myocarditis complicated by arrhythmias and multiorgan failure. This case also highlights the importance of a multidisciplinary approach, with close collaboration among intensivists, cardiologists, hematologists, and rheumatologists to optimize care. Looking ahead, there is a pressing need for predictive biomarkers and standardized treatment protocols to improve early detection and guide therapy for ICI-related myocarditis, as well as strategies for identifying and mitigating risks in vulnerable patients. Despite the challenges, timely intervention remains the best hope for improving outcomes in this devastating condition.

The prognosis of cardiac toxicities related to anthracyclins or new cancer therapies can vary significantly. Interventions to rule out other common causes (such as infectious complications and ischemia) are of importance while concomitantly discontinuing the agent and treating accordingly (Table 3). Cardiac toxicities associated to cell therapy and Bispecific monoclonal antibodies (BiTES) include arrhythmias and cardiomyopathy in the setting of cytokine release syndrome (CRS) [71,72]. These resolve as CRS improves and are not associated with poor outcomes [69]. Cardiac toxicities associated to targeted therapies (kinase inhibitors and monoclonal Abs) include QTc prolongation, atrial

Table 3
Cardiac complications associated to treatments of hematological malignancies

Toxicity/ Syndrome	Etiology	Prognosis	Treatment and comments
Cardiotoxicity	CAR-T cell therapy and bispecific monoclonal antibodies	Resolves as CRS resolves, not associated with poor outcomes.	Cardiotoxicity is mainly observed with CRS but reports of on-target off-tumor toxicities have been reported. Arrhythmias are the most common complications. Supportive care and management of CRS
	Checkpoint inhibitors	Varies according to severity: Grade I-II minimal symptoms or only elevation of cardiac markers (can resolve if treated early); Grade III and IV usually associated to poor prognosis specially if treatment is delayed. Can be associated to neuromuscular syndromes, in which case mortality can be as high as 80%.	Rule out other causes of cardiomyopathy (ischemic, infectious, etc...) concomitant with initiation of immunosuppressive therapy. Normal echocardiogram can still be associated with sudden death in patient with myocarditis.
	Targeted therapies: Kinase inhibitors and monoclonal antibodies	Many toxicities reverse with discontinuation of the agent.	Toxicities include cardiomyopathies, QTc prolongation, hypertensive emergencies, pulmonary hypertension and arrhythmias.

CAR: Chimeric Antigen Receptor, CRS: Cytokine release syndrome.

fibrillation and other dysrhythmias, hypertension(HTN), pulmonary HTN and cardiomyopathies [73]. Treatment is supportive and toxicities are usually reversible after discontinuing therapy. Menin-inhibitors, targeted agents that block the interaction of menin with KMT2A or NPM1, cause differentiation syndrome with as many as 20% of patients developing pericardial effusions [4]. As with CRS, if differentiation syndrome improves, cardiac toxicities should resolve. Many patients can continue this therapy even after developing these toxicities [74]. ICIs can cause myocarditis, pericarditis and arrhythmias [75]. Treatment needs to be urgent, even for those patients who are asymptomatic, as mortality can be high if treatment is delayed. ICIs can be restarted in patients with grade 1 myocarditis under close monitoring [75]. Discussions of prognosis in the setting of cardiotoxicities due to new cancer therapies should not be limited to the severity of the toxicity, but its reversibility and the patient’s ability to receive further therapy.

4. Approach to diagnosing and managing critical conditions

4.1. ICU admission

The increasing number of elderly patients with hematologic malignancies (HM) considered for ICU admission reflects a changing population with more comorbidities, yet a relatively good health condition [4]. ICU admission is recommended when acute organ dysfunction occurs. Triage decisions should not be solely guided by clinical assessment and physiological scores given their limitations [76,77]. Beyond the number and severity of organ dysfunction, validated scores assessing overall health and the burden of comorbidities should take precedence over age alone when triaging patients. Additionally, the eligibility for anticancer treatment holds greater weight than the characteristics of the malignancy itself. The optimal timing for ICU admission remains a subject of debate. While delayed admission is associated with higher mortality [77], the benefits of early admission remain controversial [27]. Close collaboration between intensivists and hematologists is crucial to prevent delays and facilitate informed decisions during off-hours [78]. Finally, recent advancements have improved the prognosis of allogeneic hematopoietic stem-cell transplantation recipients in the ICU [68].

4.2. The use of targeted therapy in the ICU

More than 40 targeted drugs are currently approved for clinical use. It is noteworthy that NTRK fusion inhibitors and PD-1 antagonists approval solely based on their biological function independent of a tumor of origin, has generated big changes in the clinical arena [79]. Diagnostics are based either on targeted analysis of single known mutations or using panel or whole exome/genome sequencing [80].

This development is continuously gaining momentum and brings various challenges to the ICU. Indication for a targeted drug can only be made using a relatively recent histology and in the case of panel diagnostics take time. In order to provide the relevant molecular information, it might therefore be necessary to biopsy the patients while still in the critical care unit with unfavorable conditions such as thrombocytopenia, coagulation disorders, invasive ventilation etc. Performance of diagnostics and interpretation require formal accreditation and hands-on expertise. Therefore, this treatment modality can only be offered in larger centers or in collaboration with these. Virtual molecular tumor boards nowadays offer remote consultations without the need to transfer patients. Despite the idea of being a personalized and targeted treatment these drugs have class-specific off-target toxicities among these atrial fibrillation (e.g. Ibrutinib), long QT (e.d. midostaurin), tumor lysis (antibodies, venetoclax), autoimmunity (e.g. immune checkpoint inhibitors) as well as CRS, ICANS and immune effector cell-associated hematotoxicity (ICAHT, e.g. CAR T-cells, bispecific antibodies) are typical side effects that can be identified early and managed efficiently by experienced interprofessional teams [81]. While the above

are specific for single drug classes, secondary immune deficiencies are common to most of these (Table 4).

4.3. Blood product transfusions in hematology patients

The treatment of hematological malignancies frequently induces anemia and thrombocytopenia, compounding the already high risk of anemia in these patients due to factors such as nutritional deficiencies, severe inflammation, and bone marrow infiltration. While large trials in general critically ill populations have demonstrated the safety of a restrictive red blood cell (RBC) transfusion strategy, defined by a hemoglobin threshold of 7 g/dl [82], no specific studies have evaluated transfusion strategies in critically ill hematology patients [83,84]. The Surviving Sepsis Campaign endorses a restrictive transfusion approach; however, recent studies in critically ill cancer patients, excluding those with hematological malignancies, suggest that a more liberal RBC transfusion strategy may provide greater benefits [85,86]. Outside the ICU, a recent randomized controlled trial (RCT) involving 300 hematological patients undergoing hematopoietic stem cell transplantation (HSCT) found no significant differences in clinical outcomes between restrictive and liberal RBC transfusion strategies, with the restrictive approach leading to fewer transfusions [87].

In terms of platelet transfusion thresholds, RCTs have consistently shown that therapeutic-only platelet transfusions result in a higher bleeding risk compared to prophylactic transfusions at a threshold of 10×10^9 cells/L. [88,89] Increasing the threshold above 10×10^9 cells/L has not been demonstrated to reduce bleeding rates in non-ICU patients [90]. Guidelines from the American Society of Clinical Oncology and the British Society for Haematology recommend prophylactic platelet

transfusions for counts below 10×10^9 cells/L in patients receiving active cancer therapy or undergoing HSCT [91]. For critically ill patients with hematological malignancies who often present additional bleeding risk factors—such as ongoing infection, sepsis, severe mucositis, or rapidly declining platelet counts—considering a higher transfusion threshold of 10 to 20×10^9 cells/L may be prudent. These adjustments are particularly relevant to this patient population due to the frequent presence of these risk factors in the intensive care setting. Higher platelet transfusion threshold should be considered in patients with acute promyelocytic leukemia.

4.4. Updating risk factors for mortality

Hematologic malignancies have become chronic diseases. With new therapies, people live longer and are prone to new complications or decompensations that would not have been observed otherwise. The most common reasons for ICU admission include (1) complications that are not specific but frequently observed in hematologic malignancies, that is sepsis and respiratory failure [4], the outcomes of which have improved over time [92], but remain poor [49]; (2) complications that are specific to hematologic malignancies [93]; (3) immune targeted therapy-related complications (e.g., immune check point inhibitors or CAR-T cell therapy) [19,71,94]; (4) late effects of chronic graft-versus-host disease (e.g., subsequent neoplasms) following hematopoietic cell transplantation [95]; and (5) complications related to any underlying medical condition (e.g. chronic obstructive lung disease decompensation). Risk factors of mortality of hematologic patients relate to the patients themselves, the care team and the environment. Patients factors include age, comorbidity, frailty, severity score and code status on

Table 4
Typical side effects of targeted drugs

Drug class	Example	Immune defect	Clinical correlate	Recommendation
CD20 mab	Rituximab	Limited and long-term B cell aplasia	Bacterial infections e.g. respiratory (Haemophilus, pneumococci)	Screen for Hepatitis B and treat if HBsAg (+), monitor viral load
		Early and late neutropenia	Viral infections	Monitor Immunoglobulin and substitute if low
Bcl-2 inhibitor	Venetoclax	Neutropenia	Bacterial infections grade 3 or 4 in 20%	Monitor vaccination responses and boost Monitor for infections, frequent in combinations with other immunosuppressant drugs
Bcr-abl	Imatinib, Ponatinib	Weak	Few bacterial and viral infections	Reduce 75% with azoles (CYP3A4) None
Bispecific antibody	Blinatumomab	Long-term B cell aplasia	Infections >grade 2 (25%)	Monitoring and substitution of Immunoglobulins
		Hypogammaglobulinemia	Central line infections	Monitor central line
		Neutropenia	No experience for patients with Hepatitis B, C and HIV	Test for HepB/C and HIV
BTK tyrosine kinase inhibitor	Ibrutinib	Impaired B & T cell function	Bacterial infections e.g. pneumonia, Hepatitis B virus reactivation, fungal infections	Hepatitis B screening, therapy & monitoring
		Impaired innate immunity		DDI with azoles (CAP3A4)
CAR-T cells	Axicabtagen-Cileucel	B cell aplasia	Bacterial, viral and fungal infections are frequent (non-relapse mortality due to infections: 60-80%)	Antifungal prophylaxis in long-term neutropenia or steroid treatment Monitor and substitute Ig
		Long-term neutropenia in 25%		Antibacterial and antifungal prophylaxis in high risk patients according to local standards
Immune checkpoint inhibitors	Pembrolizumab	Unknown	Infections, especially when irAE appear and need to be treated with steroids	Monitor and consider prophylaxis (e.g. PJP) if irAE are treated with steroids > 4 weeks
Jak-Stat inhibitors	Ruxolitinib		Discreetly increase bacterial, viral and fungal infections	Consider PJP prophylaxis if CD4<200 or steroids Hepatitis B virus monitoring & treatment Screening for opportunistic infections & PML

DDI: Drug-drug interactions; HIV: Human Immunodeficiency Virus; irAE: Immune-related adverse events; mab: monoclonal antibody; PJP: Pneumocystis jirovecii pneumonia; PML: Progressive Multifocal Leukoencephalopathy.

admission [22]. Care team factors include familiarity with hematologic patients and availability of a hematologist, and more broadly speaking of a multidisciplinary team (e.g., clinical pharmacist, or a transfusion specialist) [78,96]. Environmental factors include the presence of a structured outreach emergency system, with early recognition, and early rescue by specifically trained rapid response teams, bed availability, early ICU admission and definite diagnosis [97], use of telemedicine such as the telehealth Serious Illness Care Program [98] and ongoing goal of care discussion with patients and their family [46]. Some specific therapies (e.g. aminoglycosides, colony stimulating factors, extracorporeal circulation) are still being investigated. The focus of care should not only be on mortality (e.g., 30-day mortality) but on the cure of the underlying malignancy and quality of life (e.g., ability to returning home) (BHPC 2023-24 (usnews.com)).

4.5. Time-limited trials

Identifying patients who will benefit from life-sustaining therapies (LST) might be challenging because of uncertainty regarding the reversibility of critical illness and/or the patient's preferences. ICU time-limited trial (TLT) have been proposed to face these situations of doubt, avoiding burdensome and inappropriate emergent decisions, as well as upholding of excessive cares that affect not only the patient, but also family and caregivers [76,99–101].

Collegiality and communication are two cornerstones of TLT. Multidisciplinary discussion, involving at least intensivists and hematologists, should lead to a precise definition of TLT modalities including acceptable life sustaining therapies according to patient's preferences, duration of TLT, criteria to evaluate response to treatment and potential decisions to be made at the end of the trial. TLT plan is shared with the patient, if possible, and/or her/his surrogate(s) and with all caregivers. Meeting are regularly scheduled to reappraise goals of care according to patient's evolution [102]. For instance, decision making should also be guided by the onset of additional organ failures that carry a more adverse outcome than merely prolonging the initial failure requiring critical care.

Optimal duration of ICU trial depends on multiple factors (patient's frailty, severity and etiology of organ failures, expected treatment response time) and should be personalized [76,103].

4.6. Setting realistic goals of care and maintaining hope while avoiding unnecessary suffering

With advances in the care of critically ill patients who have hematologic malignancy, they and their families frequently come to the ICU with high expectations of intensive therapies and hope [104]. Many also experience fear and uncertainty that is disturbing and sometimes traumatic in the short or longer term [105–107]. Patients, families and clinicians all desire the delivery of critical care that is aligned with the patients' goals, but may not communicate as needed to optimize goal concordance. To establish meaningful goals of care, the framework for communication must be a mutual exchange in which patients or their surrogates are supported to articulate the patient's core health-related values while the interprofessional/interdisciplinary team provides clinical expertise and empathy as a context for shared decision-making [108–110]. Discussion of hopes can be especially rich, revealing, and productive when the patient is asked not only, 'what do you hope for?', which may not be possible, but to bring forward achievable goals that are also important to pursue [111]. Understanding who patient is as a person, and what matters most to this unique individual, is the foundation for person-centered communication, care, and decision-making that affirms personhood and realistic hopes, maximizing appropriate use of intensive care and minimizing distress for patients, their loved ones, and clinicians [110–113]. The finding that ICU survivors have the same quality of life of hematology patients never admitted to the ICU or to non-hematology critically ill patients offers new perspectives into ICU

management of these patients [114].

4.7. End-of-life care and symptom management

ICU patients experience many adverse symptoms such as pain, thirst, anxiousness, tiredness, and shortness of breath from their diseases or ICU treatments and procedures. This symptom burden in high-risk patients was identified years ago [115,116], but continues today [117]. Multiple symptoms have been identified in critically ill hematologic malignancy patients, including pain and dyspnea [118]. Recent critical care organizational initiatives stressed that attention be given to symptom assessment in patients with advanced medical illnesses [41,118]. Systematic symptom assessment precedes intentional symptom management. Interventions tailored to specific symptoms or discomforts improved many discomforts of ICU patients [8] and lowered rates of post-traumatic stress disorder in survivors [119]. Methods to palliate pain, dyspnea, and thirst have been identified [120], yet methods to alleviate these and other ICU patient symptoms must continue to be explored.

5. Infection management

5.1. Immune defenses and infection types in immunodeficient patients

The approach to diagnosing and managing suspected infections in hematology patients in the ICU is fundamentally guided by the type and degree of immunosuppression, which varies significantly depending on the underlying hematologic condition and prior treatments [2,121]. These patients are particularly vulnerable to infection but represent a heterogeneous population with unique pathogen risks determined by their immunological deficits.

For patients with short-term neutropenia (fewer than seven days), the primary risk is bacterial infections [122]. In contrast, prolonged neutropenia, commonly seen after induction therapy for acute myeloid leukemia, predisposes patients to invasive fungal infections, such as those caused by *Aspergillus* species or *Mucorales* [123,124]. Impaired humoral immunity, as seen in conditions like asplenia or hypogammaglobulinemia (e.g., in multiple myeloma, especially after treatment with bispecific antibodies), increases susceptibility to life-threatening infections by encapsulated bacteria, such as *Streptococcus pneumoniae* [125].

Patients with T-cell deficiencies, often associated with lymphoid malignancies or therapies involving T-cell depletion (e.g., fludarabine), are at increased risk of fungal infections like *Pneumocystis jirovecii* or *Cryptococcus*, as well as herpesvirus infections, including CMV and HSV [126]. Dysfunction of monocytes and macrophages, as occurs in diseases like hairy cell leukemia, predisposes individuals to infections by intracellular pathogens such as mycobacteria and *Legionella* species [127]. Additionally, the use of novel therapies like ibrutinib, a Bruton's tyrosine kinase (BTK) inhibitor, has been associated with a heightened risk of invasive fungal infections, particularly cerebral aspergillosis in the context of concomitant steroid therapy [128].

5.2. Antibiotic management

In critically ill hematological patients, particularly those with neutropenia or multiple myeloma and hypogammaglobulinemia, bacterial infections are associated with mortality [129]. Prompt initiation of antibiotic therapy is critical in this setting, as delays of even one hour in starting appropriate treatment after the detection of organ dysfunction can result in a tenfold increase in adjusted mortality among neutropenic patients [43,122]. This urgency remains crucial even in clinical conditions that can mimic sepsis, such as Cytokine Release Syndrome, given the high prevalence and severity of bacterial infections in this population [130,131].

Empirical antibiotic therapy should include a beta-lactam with

activity against *Pseudomonas aeruginosa* as the preferred initial choice. For patients with sepsis or septic shock, the addition of an agent effective against Methicillin-Resistant *Staphylococcus aureus* (MRSA), such as glycopeptides, daptomycin, or linezolid, is recommended [129,132]. Although the routine use of combination therapy with aminoglycosides remains a standard practice for critically ill patients in many centers, its benefits and tolerability continue to be debated [133]. This broad-spectrum empirical therapy should be reassessed and de-escalated rapidly based on clinical and microbiological findings. Glycopeptides, daptomycin, or linezolid can typically be implemented as second line therapy in non-ICU patients [134], or discontinued by day three in patients without evidence of resistant *Staphylococcus aureus* or beta-hemolytic *Streptococcus* infections. Aminoglycosides are usually stopped after microbiological documentation or, in the absence of such documentation, by day three. De-escalation of the initial beta-lactam is considered safe once microbiological results are available and guide therapy appropriately [135].

For neutropenic patients without organ dysfunction or clinically documented infection, empirical antibiotics may be discontinued after three days of sustained apyrexia. However, this approach has not been tested in critically ill patients, where antibiotics are generally required throughout the duration of neutropenia [136,137]. For patients with documented infections, antibiotics are typically continued until the usual duration of therapy has been completed and neutropenia has resolved. In cases without documentation, antibiotics are often discontinued on the second day after neutropenia recovery [129,132]. Ongoing research aims to clarify the impact of de-escalation strategies and the role of aminoglycosides in critically ill neutropenic patients. Additionally, future studies are needed to evaluate the feasibility and safety of early antibiotic discontinuation in this high-risk population. These efforts could help optimize antibiotic stewardship while ensuring the best possible outcomes for critically ill hematological patients.

5.3. Invasive fungal infections

Invasive fungal infections (IFIs) in immunocompromised critically ill patients are a significant challenge, driven by antibiotic overuse, which disrupts the microbiota, leading to dysbiosis and secondary infections [138,139]. This ecological imbalance has contributed to increased non-albicans candidemia, which is often resistant to conventional antifungals and complicates management [140,141]. The development of advanced molecular diagnostics, including next-generation sequencing and polymerase chain reaction-based assays, enables rapid and precise pathogen identification, facilitating earlier detection and targeted therapy [142]. Simultaneously, novel antifungal agents, such as new azoles and echinocandins, are expanding treatment options, addressing resistance, and improving outcomes, especially at a time of generalized anti-aspergillus prophylaxis [143,144].

5.4. Antimicrobial stewardship

In immunosuppressed patients, the challenges of antimicrobial stewardship are heightened due to their vulnerability to opportunistic infections [135]. Overuse of broad-spectrum antibiotics and empirical antifungal therapies in this population not only fosters resistance but also increases the risk of breakthrough infections and toxicity. Effective stewardship programs in immunosuppressed patients require a tailored approach, emphasizing early, aggressive diagnostic efforts combined with precision medicine to guide therapy [145]. Advanced molecular diagnostics, such as T2Candida panel (T2 magnetic resonance technology) and beta-D-glucan assays, can aid in early detection, allowing targeted antifungal therapies and minimizing unnecessary drug exposure. Additionally, novel antifungal agents, such as isavuconazole and liposomal amphotericin B, have shown promise in addressing resistance patterns and improving outcomes in these high-risk patients [146]. Multidisciplinary collaboration is essential to integrate these

advancements into clinical practice, ensuring the optimal management of IFIs while balancing the risks and benefits of treatment in immunosuppressed critically ill patients.

5.5. Opportunistic infections

The main risk factors for opportunistic infections in hematologic patients are treatments that impair T-cell immunity, such as corticosteroids, acute lymphoblastic leukemia therapies, and hematopoietic stem cell transplantation (HSCT). Prior exposure to the microorganism and the absence of prophylaxis are the most critical factors influencing the likelihood of infection.

Pneumocystis jirovecii pneumonia (PJP) is a significant cause of acute respiratory failure, occurring in up to 12% of hematologic patients admitted to the ICU, with mortality rates reaching as high as 50% [17,147]. Advances in noninvasive diagnostic strategies, such as positive quantitative PCR in induced sputum combined with elevated serum (1,3)- β -D-glucan levels, have reduced the reliance on bronchoscopy, minimizing procedure-related complications [148,149].

The introduction of antiviral prophylaxis, including acyclovir for herpes simplex virus and letermovir for cytomegalovirus (CMV) prophylaxis in seropositive HSCT recipients, has substantially reduced the incidence of viral reactivation [150]. Despite these advances, viral reactivation remains common in the ICU and can lead to severe organ disease [151,152]. CMV pneumonitis, the most frequent CMV-related end-organ disease in critically ill hematology patients, is associated with high mortality rates [153]. Other viruses, such as adenovirus and human herpesvirus-6 (HHV6), lack established prophylactic measures. HHV6, notably the leading cause of encephalitis in allogeneic HSCT recipients, can progress to multiple organ involvement, complicating management in this vulnerable population [154,155].

Less common opportunistic infections, such as toxoplasmosis and mycobacterial infections—including both tuberculosis and non-tuberculous mycobacteria—pose additional diagnostic and therapeutic challenges. These infections are particularly problematic due to potential drug interactions and the need for prolonged, complex treatment regimens. Together, these opportunistic infections underscore the importance of robust prophylactic strategies, early diagnosis, and tailored management to improve outcomes in critically ill hematologic patients.

5.6. Lessons from general and specific guidelines on sepsis management

Cancer patients face a significantly heightened risk of severe infections due to various immunological impairments, including treatment-associated neutropenia, lymphopenia, and dysfunction of T and B cells. Sepsis and septic shock are among the leading causes of mortality in this population [122,156]. The diverse range of cancer treatments and their effects on the immune system can result in atypical infection responses and distinct septic processes, making early recognition of unclear or nonspecific clinical symptoms critical. Prompt identification and intervention are essential to ensure timely management, as enhanced sepsis care has the potential to reduce mortality associated with cancer therapies.

The management of septic shock in cancer patients should follow standard evidence-based guidelines for sepsis and septic shock, emphasizing the importance of early resuscitation, infection source control, and appropriate antimicrobial therapy [82,156]. Timely admission to the ICU is particularly crucial for cancer patients with sepsis, as delays have been consistently linked to worse outcomes. Decision-making regarding ICU admission should extend beyond rigid, predefined criteria, as these can vary widely between institutions, cultural settings, and countries [157,158]. Instead, a more flexible, patient-centered approach that considers the individual's clinical status, prognosis, and therapeutic goals is recommended to optimize outcomes for this vulnerable population.

6. Future directions and research needs

The intersection of critical care and hematological malignancies demands ongoing innovation to address existing challenges and improve outcomes. Future research should prioritize several key areas that are detailed in Table 5. Future research must adopt a multidisciplinary approach, ensuring that advancements in hematology, oncology, and critical care are integrated to holistically improve patient outcomes.

7. Conclusion

Critical care management for patients with hematological malignancies has undergone remarkable advances, offering new opportunities to improve survival and quality of life. However, these achievements bring complexities that demand a nuanced and multidisciplinary approach to care. This manuscript highlights the evolving epidemiology of hematological malignancies, emphasizing the critical role of early ICU admission, tailored therapeutic strategies, and effective collaboration among healthcare teams. The challenges of managing complications, including infections, coagulopathies, and organ dysfunction, are compounded by the growing use of innovative therapies such as CAR-T cells and bispecific antibodies. Moving forward, the integration of precision medicine, digital health tools, and patient-centered care frameworks will be pivotal in advancing the field. Simultaneously, research efforts must address gaps in knowledge, optimize resource utilization, and ensure equitable access to life-saving interventions. Ultimately, the goal remains clear: to provide care that not only extends survival but also upholds the dignity, autonomy, and quality of life of patients. By building on the lessons of the past and embracing the possibilities of the future, critical care for hematological malignancies can continue to evolve as a model of innovation and compassion.

Practice points

- Rising ICU admissions for hematologic malignancies
- Key drivers are respiratory failure, infections, sepsis, treatment toxicity
- Complex cases (e.g., leukemic infiltrates, TLS, opportunistic infections) increase ICU demands

Table 5
Future directions and research needs.

1. Clarifying diagnostic workup Diagnosis of CRS and other therapy-related cytokine storms and HLH is still based on a combination of criteria and thus remain subject to the physician's judgment and experience. Formally establishing the diagnosis and/or ruling out infection can be particularly challenging as these entities are not mutually exclusive and may trigger one another (e.g., CRS and sepsis, or HLH triggered by both infections and the underlying malignancy).
2. Establishing the risk/benefit ratio of several procedures Organ failure that complicates the procedures and the interpretation of many biological parameters is a second hit for these frail patients.
3. Improving the level of evidence for treatment-related toxicities We lack robust randomized clinical trials to address this issue and standardize the treatments of these toxicities.
4. Precision Medicine in Critical Care The integration of genomic, transcriptomic, and proteomic data to stratify patients based on their unique disease and treatment profiles could enable tailored interventions
5. Biomarker-driven approaches To be evaluated for early detection of complications such as sepsis, CRS, and invasive fungal infections hold promise for reducing morbidity and mortality
6. Optimizing Treatment Modalities Continued exploration of novel therapies, including next-generation CAR-T cells, bispecific antibodies, and targeted agents, should extend beyond efficacy to assess their safety profiles in critically ill patients. Identifying strategies to mitigate treatment-related toxicities (cardiotoxicity, neurotoxicity, coagulopathies) is essential
7. Advancing ICU Protocols Multidisciplinary collaboration between hematologists and intensivists must be bolstered through standardized protocols and decision-making frameworks. Research should investigate the timing and intensity of ICU interventions, including early admission policies and time-limited trials, to maximize patient outcomes
8. Patient-Centered Outcomes Beyond survival, the focus should shift toward assessing long-term quality of life, functional recovery, and psychological well-being. Studies evaluating post-ICU care, including rehabilitation and mental health support, will be critical in addressing the unique needs of this population
9. Digital Health and Artificial Intelligence Leveraging telemedicine and AI for early warning systems, predictive analytics, and remote patient monitoring could transform care delivery. Digital tools can also enhance antimicrobial stewardship and infection management, particularly in resource-limited settings
10. Global and Equity-Focused Initiatives: Recognizing the disparities in access to advanced therapies and ICU care, international collaborations should aim to develop cost-effective solutions and expand access to underserved populations.

AI: Artificial intelligence; CAR: Chimeric antigen receptor; ICU: Intensive Care Unit.

- New therapies (CAR-T, bispecifics, checkpoint inhibitors) offer promise but pose toxicity risks
- Survival rates are improving
- Early ICU, teamwork, and precision medicine boost outcomes
- Clearer guidance are needed for time-limited trials and goal-driven care

Research agenda

- **Precision Medicine in Critical Care:** The integration of genomic, transcriptomic, and proteomic data to stratify patients based on their unique disease and treatment profiles could enable tailored interventions for critically ill hematology patients
- **Biomarker-driven approaches:** To be evaluated for early detection of complications such as sepsis, cytokine release syndrome, and invasive fungal infections hold promise for reducing morbidity and mortality
- **Optimizing Treatment Modalities:** Continued exploration of novel therapies, including next-generation CAR-T cells, bispecific antibodies, and targeted agents, should extend beyond efficacy to assess their safety profiles in critically ill patients. Identifying strategies to mitigate treatment-related toxicities, including cardiotoxicity, neurotoxicity, and coagulopathies, is essential
- **Advancing ICU Protocols:** Multidisciplinary collaboration between hematologists and intensivists must be bolstered through standardized protocols and decision-making frameworks. Research should investigate the timing and intensity of ICU interventions, including early admission policies and time-limited trials, to maximize patient outcomes
- **Patient-Centered Outcomes:** Beyond survival, the focus should shift toward assessing long-term quality of life, functional recovery, and psychological well-being. Studies evaluating post-ICU care, including rehabilitation and mental health support, will be critical in addressing the unique needs of this population
- **Digital Health and Artificial Intelligence:** Leveraging telemedicine and AI for early warning systems, predictive analytics, and remote patient monitoring could transform care delivery. Digital tools can also enhance antimicrobial stewardship and infection management, particularly in resource-limited settings

- **Global and Equity-Focused Initiatives:** Recognizing the disparities in access to advanced therapies and ICU care, international collaborations should aim to develop cost-effective solutions and expand access to underserved populations.

References

- [1] Bray F, Laversanne M, Sung H, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 2024;74(3):229–63.
- [2] Azoulay E, Mokart D, Kouatchet A, Demoule A, Lemiale V. Acute respiratory failure in immunocompromised adults. *Lancet Respir Med* 2019;7(2):173–86.
- [3] Choudhry A, DeLoughery TG. Bleeding and thrombosis in acute promyelocytic leukemia. *Am J Hematol* 2012;87(6):596–603.
- [4] Azoulay E, Mokart D, Pène F, et al. Outcomes of critically ill patients with hematologic malignancies: prospective multicenter data from France and Belgium—a groupe de recherche respiratoire en réanimation onco-hématologique study. *J Clin Oncol Off J Am Soc Clin Oncol* 2013;31(22):2810–8.
- [5] Joseph A, Harel S, Venot M, et al. Renal recovery after severe acute kidney injury in critically ill myeloma patients: a retrospective study. *Clin Kidney J* 2018;11:20–5.
- [6] Carini FC, Angriman F, Scales DC, et al. Venous thromboembolism in critically ill adult patients with hematologic malignancy: a population-based cohort study. *Intensive Care Med* 2024;50(2):222–33.
- [7] Abaza Y, McMahon C, Garcia JS. Advancements and challenges in the treatment of AML. *Am Soc Clin Oncol Educ Book* 2024;44(3):e438662.
- [8] Venditti A, Hou J-Z, Fenaux P, et al. Outcomes in chemotherapy-ineligible elderly patients with newly diagnosed acute myeloid leukemia treated with venetoclax plus azacitidine: a pooled analysis. *Blood* 2023;142(Supplement 1):2886.
- [9] Mamez A-C, Raffoux E, Chevret S, et al. Pre-treatment with oral hydroxyurea prior to intensive chemotherapy improves early survival of patients with high hyperleukocytosis in acute myeloid leukemia. *Leuk Lymphoma* 2016;57(10):2281–8.
- [10] Cerrano M, Chevret S, Raffoux E, et al. Benefits of dexamethasone on early outcomes in patients with acute myeloid leukemia with hyperleukocytosis: a propensity score matched analysis. *Ann Hematol* 2023;102(4):761–8.
- [11] Gebru MT, Atkinson JM, Young MM, et al. Glucocorticoids enhance the antileukemic activity of FLT3 inhibitors in FLT3-mutant acute myeloid leukemia. *Blood* 2020;136(9):1067–79.
- [12] Ferreyro BL, Scales DC, Wunsch H, et al. Critical illness in patients with hematologic malignancy: a population-based cohort study. *Intensive Care Med* 2021;47(10):1104–14.
- [13] Dumas G, Biard L, Givel C, et al. Intensive care unit admission in patients with T cell lymphomas: clinical features and outcome. *Ann Hematol* 2019;98:195–203.
- [14] Wohlfarth P, Carlström A, Staudinger T, et al. Incidence of intensive care unit admission, outcome and post intensive care survival in patients with diffuse large B-cell lymphoma. *Leuk Lymphoma* 2016;57(8):1831–8.
- [15] Abdel-Nabey M, Chaba A, Serre J, et al. Tumor lysis syndrome, acute kidney injury and disease-free survival in critically ill patients requiring urgent chemotherapy. *Ann Intensive Care* 2022;12:15.
- [16] Darmon M, Vincent F, Camous L, et al. Tumour lysis syndrome and acute kidney injury in high-risk haematology patients in the rasburicase era. A prospective multicentre study from the Groupe de Recherche en Réanimation Respiratoire et Onco-Hématologique. *Br J Haematol* 2013;162:489–97.
- [17] Azoulay E, Roux A, Vincent F, et al. A multivariable prediction model for pneumocystis jirovecii pneumonia in hematology patients with acute respiratory failure. *Am J Respir Crit Care Med* 2018;198:1519–26.
- [18] Maertens J, Cesaro S, Maschmeyer G, et al. ECIL guidelines for preventing Pneumocystis jirovecii pneumonia in patients with hematological malignancies and stem cell transplant recipients. *J Antimicrob Chemother* 2016;71(9):2397–404.
- [19] Azoulay É, Castro P, Maamar A, et al. Outcomes in patients treated with chimeric antigen receptor T-cell therapy who were admitted to intensive care (CARTTAS): an international, multicentre, observational cohort study. *Lancet Haematol* 2021;8:e355–64.
- [20] Gutierrez C, Brown ART, Herr MM, et al. The chimeric antigen receptor-intensive care unit (CAR-ICU) initiative: surveying intensive care unit practices in the management of CAR T-cell associated toxicities. *J Crit Care* 2020;58:58–64.
- [21] Siegel RL, Giaquinto AN, Jemal A. Cancer statistics, 2024. *CA Cancer J Clin* 2024;74(1):12–49.
- [22] Munshi L, Dumas G, Rochweg B, et al. Long-term survival and functional outcomes of critically ill patients with hematologic malignancies: a Canadian multicenter prospective study. *Intensive Care Med* 2024;50(4):561–72.
- [23] Nassar AP, Archanjo LVF, Ranzani OT, et al. Characteristics and outcomes of autologous hematopoietic stem cell transplant recipients admitted to intensive care units: a multicenter study. *J Crit Care* 2022;71:154077.
- [24] Dimopoulos MA, Merlini G, Bridoux F, et al. Management of multiple myeloma-related renal impairment: recommendations from the International Myeloma Working Group. *Lancet Oncol* 2023;24(7):e293–311.
- [25] Bridoux F, Carron P-L, Pegourie B, et al. Effect of high-cutoff hemodialysis vs conventional hemodialysis on hemodialysis independence among patients with myeloma cast nephropathy: a randomized clinical trial. *JAMA* 2017;318(21):2099–110.
- [26] Lancman G, Parsa K, Kotlarz K, et al. IVIg use associated with ten-fold reduction of serious infections in multiple myeloma patients treated with anti-bcma bispecific antibodies. *Blood Cancer Discov* 2023;4(6):440–51.
- [27] Hourmant Y, Kouatchet A, López R, et al. Impact of early ICU admission for critically ill cancer patients: post-hoc analysis of a prospective multicenter multinational dataset. *J Crit Care* 2021;62:6–11.
- [28] Bauer P, Chevret S, Yadav H, et al. Diagnosis and outcome of acute respiratory failure in immunocompromised patients after bronchoscopy. *Eur Respir J* 2019;54.
- [29] Azoulay E, Mokart D, Lambert J, et al. Diagnostic strategy for hematology and oncology patients with acute respiratory failure: randomized controlled trial. *Am J Respir Crit Care Med* 2010;182(8):1038–46.
- [30] Legoff J, Zucman N, Lemiale V, et al. Clinical significance of upper airway virus detection in critically ill hematology patients. *Am J Respir Crit Care Med* 2019;199:518–28.
- [31] Dumas G, Lemiale V, Rath N, et al. Survival in immunocompromised patients ultimately requiring invasive mechanical ventilation: a pooled individual patient data analysis. *Am J Respir Crit Care Med* 2021;204:187–96.
- [32] Stahl M, Shallis RM, Wei W, et al. Management of hyperleukocytosis and impact of leukapheresis among patients with acute myeloid leukemia (AML) on short- and long-term clinical outcomes: a large, retrospective, multicenter, international study. *Leukemia* 2020;34(12):3149–60.
- [33] Porcu P, Danielson CF, Orazi A, et al. Therapeutic leukapheresis in hyperleukocytic leukaemias: lack of correlation between degree of cyto-reduction and early mortality rate. *Br J Haematol* 1997;98(2):433–6.
- [34] Bewersdorf JP, Giri S, Tallman MS, Zeidan AM, Stahl M. Leukapheresis for the management of hyperleukocytosis in acute myeloid leukemia—a systematic review and meta-analysis. *Transfusion (Paris)* 2020;60(10):2360–9.
- [35] Rinaldi I, Sutandyo N, Winston K. Comparison of early mortality between leukapheresis and non-leukapheresis in adult acute myeloid leukemia patients with hyperleukocytosis: a systematic review and meta-analysis. *Hematol Amst Neth* 2022;27(1):141–9.
- [36] Green RA, Nichols NJ. Pulmonary Involvement in Leukemia. *Am Rev Respir Dis* 1959;80(6):833–44.
- [37] Nathan DJ, Sanders M. Manifestations of acute leukemia in the parenchyma of the lung. *N Engl J Med* 1955;252(19):797–801.
- [38] Döhner H, Wei AH, Appelbaum FR, et al. Diagnosis and management of AML in adults: 2022 recommendations from an international expert panel on behalf of the ELN. *Blood* 2022;140(12):1345–77.
- [39] Kovalski R, Hansen-Flaschen J, Lodato RF, Pietra GG. Localized leukemic pulmonary infiltrates. Diagnosis by bronchoscopy and resolution with therapy. *Chest* 1990;97(3):674–8.
- [40] Dombret H, Hunault M, Faucher C, Dombret MC, Degos L. Acute lysis pneumopathy after chemotherapy for acute myelomonocytic leukemia with abnormal marrow eosinophils. *Cancer* 1992;69(6):1356–61.
- [41] Grasselli G, Calfee CS, Camporota L, et al. ESICM guidelines on acute respiratory distress syndrome: definition, phenotyping and respiratory support strategies. *Intensive Care Med* 2023. *Intensive Care Med* 2023;49(7):727–59. <https://doi.org/10.1007/s00134-023-07050-7>. Epub 2023 Jun 16.
- [42] Demoule A, Antonelli M, Schellongowski P, et al. Respiratory mechanics and outcomes in immunocompromised patients with ARDS: a secondary analysis of the EFRAM study. *Chest* 2020;158(5):1947–57.
- [43] Azoulay E, Lemiale V, Mokart D, et al. Acute respiratory distress syndrome in patients with malignancies. *Intensive Care Med* 2014;40(8):1106–14.
- [44] Bris P-N, Pauly V, Orleans V, et al. Acute respiratory distress syndrome in patients with hematological malignancies: a one-year retrospective nationwide cohort study. *Ann Intensive Care* 2024;14(1):141.
- [45] Cortegiani A, Madotto F, Gregoretti C, et al. Immunocompromised patients with acute respiratory distress syndrome: secondary analysis of the LUNG SAFE database. *Crit Care Lond Engl* 2018;22(1):157.
- [46] Nates JL, Pène F, Darmon M, et al. Septic shock in the immunocompromised cancer patient: a narrative review. *Crit Care Lond Engl* 2024;28(1):285.
- [47] Park K-H, Jung YJ, Lee HJ, et al. Impact of multidrug resistance on outcomes in hematologic cancer patients with bacterial bloodstream infections. *Sci Rep* 2024;14(1):15622.
- [48] Pène F, Percheron S, Lemiale V, et al. Temporal changes in management and outcome of septic shock in patients with malignancies in the intensive care unit. *Crit Care Med* 2008;36(3):690–6.
- [49] Manjappachar NK, Cuenca JA, Ramirez CM, et al. Outcomes and predictors of 28-day mortality in patients with hematologic malignancies and septic shock defined by sepsis-3 criteria. *J Natl Compr Cancer Netw* 2022;20(1):45–53.
- [50] Mokart D, Darmon M, Resche-Rigon M, et al. Prognosis of neutropenic patients admitted to the intensive care unit. *Intensive Care Med* 2015;41(2):296–303.
- [51] Hess JR, Trachtenberg FL, Assmann SF, et al. Clinical and laboratory correlates of platelet alloimmunization and refractoriness in the PLADO trial. *Vox Sang* 2016;111(3):281–91.
- [52] Baron E, Charpentier J, François A, et al. Post-transfusion platelet responses in critically ill cancer patients with hypoproliferative thrombocytopenia. *Transfusion (Paris)* 2020;60(2):275–84.
- [53] Gernsheimer TB, Brown SP, Triulzi DJ, et al. Prophylactic tranexamic acid in patients with hematologic malignancy: a placebo-controlled, randomized clinical trial. *Blood* 2022;140(11):1254–62.
- [54] Panch SR, Guo L, Vassallo R. Platelet transfusion refractoriness due to HLA alloimmunization: Evolving paradigms in mechanisms and management. *Blood Rev* 2023;62:101135.

- [55] Wang T-F, Makar RS, Antic D, et al. Management of hemostatic complications in acute leukemia: guidance from the SSC of the ISTH. *J Thromb Haemost* 2020;18(12):3174–83.
- [56] Lafarge A, Cheadle D, Whiting L, et al. Management of hematological patients requiring emergency chemotherapy in the intensive care unit. *Intensive Care Med* 2024;50(6):849–60.
- [57] Fukatsu M, Ikezoe T. Cancer-associated thrombosis in hematologic malignancies. *Int J Hematol* 2024;119(5):516–25.
- [58] Iba T, Levy JH, Warkentin TE, et al. Diagnosis and management of sepsis-induced coagulopathy and disseminated intravascular coagulation. *J Thromb Haemost* 2019;17(11):1989–94.
- [59] Squizzato A, Hunt BJ, Kinasevitz GT, et al. Supportive management strategies for disseminated intravascular coagulation. An international consensus. *Thromb Haemost* 2016;115(5):896–904.
- [60] Valade S, Monseu G, Mariotte E, Darmon M. Diagnostic performance of hemophagocytic lymphohistiocytosis criteria and HScore in critically ill patients with severe hemophagocytic syndrome. *Crit Care Med* 2021;49(9):e874–9.
- [61] Dupont T, Darmon M, Mariotte E, et al. Etoposide treatment in secondary hemophagocytic syndrome: impact on healthcare-associated infections and survival. *Ann Intensive Care* 2022;12:101.
- [62] Locatelli F, Jordan MB, Allen C, et al. Emapalumab in children with primary hemophagocytic lymphohistiocytosis. *N Engl J Med* 2020;382(19):1811–22.
- [63] Maschalidi S, Sepulveda FE, Garrigue A, Fischer A, de Saint Basile G. Therapeutic effect of JAK1/2 blockade on the manifestations of hemophagocytic lymphohistiocytosis in mice. *Blood* 2016;128(1):60–71.
- [64] Ahmed A, Merrill SA, Alsawah F, et al. Ruxolitinib in adult patients with secondary haemophagocytic lymphohistiocytosis: an open-label, single-centre, pilot trial. *Lancet Haematol* 2019;6(12):e630–7.
- [65] Gertz MA. Acute hyperviscosity: syndromes and management. *Blood* 2018;132(13):1379–85.
- [66] Azoulay É, Soares M, Lengliné É. Preempting critical care services for patients with hematological malignancies. *Intensive Care Med* 2021;47:1140–3.
- [67] Lueck C, Stadler M, Koenecke C, et al. Improved short- and long-term outcome of allogeneic stem cell recipients admitted to the intensive care unit: a retrospective longitudinal analysis of 942 patients. *Intensive Care Med* 2018;44(9):1483–92.
- [68] Lafarge A, Dupont T, Canet E, et al. Outcomes in critically ill allogeneic hematopoietic stem-cell transplantation recipients. *Am J Respir Crit Care Med* 2024;210(8):1017–24.
- [69] Pichereau C, Lengliné É, Valade S, et al. Trajectories of acute graft-versus-host disease and mortality in critically ill allogeneic-hematopoietic stem cell recipients: the Allo-GRRR-OH score. *Bone Marrow Transplant* 2020;55:1966–74.
- [70] Orvain C, Beloncle F, Hamel J-F, et al. Allogeneic stem cell transplantation recipients requiring intensive care: time is of the essence. *Ann Hematol* 2018;97(9):1601–9.
- [71] Gutierrez C, Neilan TG, Grover NS. How I approach optimization of patients at risk of cardiac and pulmonary complications after CAR T-cell therapy. *Blood* 2023;141(20):2452–9.
- [72] Shimabukuro-Vornhagen A, Böll B, Schellongowski P, et al. Critical care management of chimeric antigen receptor T-cell therapy recipients. *Cancer J Clin* 2022;72(1):78–93.
- [73] Dy GK, Adjei AA. Understanding, recognizing, and managing toxicities of targeted anticancer therapies. *CA Cancer J Clin* 2013;63(4):249–79.
- [74] Fathi AT, DiNardo CD, Kline I, et al. Differentiation syndrome associated with enasidenib, a selective inhibitor of mutant isocitrate dehydrogenase 2: analysis of a phase 1/2 study. *JAMA Oncol* 2018;4(8):1106–10.
- [75] Palaskas N, Lopez-Mattei J, Durand JB, Ilescu C, Deswal A. Immune checkpoint inhibitor myocarditis: pathophysiological characteristics, diagnosis, and treatment. *J Am Heart Assoc* 2020;9(2):e013757.
- [76] Lecuyer L, Chevret S, Thiery G, et al. The ICU trial: a new admission policy for cancer patients requiring mechanical ventilation. *Crit Care Med* 2007;35(3):808–14.
- [77] Thiéry G, Azoulay E, Darmon M, et al. Outcome of cancer patients considered for intensive care unit admission: a hospital-wide prospective study. *J Clin Oncol* 2005;23:4406–13.
- [78] Soares M, Bozza FA, Azevedo LCP, et al. Effects of organizational characteristics on outcomes and resource use in patients with cancer admitted to intensive care units. *J Clin Oncol Off J Am Soc Clin Oncol* 2016;34(27):3315–24.
- [79] Mateo J, Steuten L, Aftimos P, et al. Delivering precision oncology to patients with cancer. *Nat Med* 2022;28(4):658–65.
- [80] Mosele F, Remon J, Mateo J, et al. Recommendations for the use of next-generation sequencing (NGS) for patients with metastatic cancers: a report from the ESMO Precision Medicine Working Group. *Ann Oncol Off J Eur Soc Med Oncol* 2020;31(11):1491–505.
- [81] Casolino R, Beer PA, Chakravarty D, et al. Interpreting and integrating genomic tests results in clinical cancer care: overview and practical guidance. *CA Cancer J Clin* 2024;74(3):264–85.
- [82] Evans L, Rhodes A, Alhazzani W, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2021. *Intensive Care Med* 2021;47(11):1181–247.
- [83] Holst LB, Haase N, Wetterslev J, et al. Lower versus higher hemoglobin threshold for transfusion in septic shock. *N Engl J Med* 2014;371(15):1381–91.
- [84] Hébert PC, Wells G, Blajchman MA, et al. A multicenter, randomized, controlled clinical trial of transfusion requirements in critical care. Transfusion Requirements in Critical Care Investigators, Canadian Critical Care Trials Group. *N Engl J Med* 1999;340(6):409–17.
- [85] de Almeida JP, Vincent J-L, Galas FRBG, et al. Transfusion requirements in surgical oncology patients: a prospective, randomized controlled trial. *Anesthesiology* 2015;122(1):29–38.
- [86] Bergamin FS, Almeida JP, Landoni G, et al. Liberal versus restrictive transfusion strategy in critically ill oncologic patients: the transfusion requirements in critically ill oncologic patients randomized controlled trial. *Crit Care Med* 2017;45(5):766–73.
- [87] Tay J, Allan DS, Chatelain E, et al. Liberal versus restrictive red blood cell transfusion thresholds in hematopoietic cell transplantation: a randomized, open label, phase III, noninferiority trial. *J Clin Oncol Off J Am Soc Clin Oncol* 2020;38(13):1463–73.
- [88] Stanworth SJ, Estcourt LJ, Powter G, et al. A no-prophylaxis platelet-transfusion strategy for hematologic cancers. *N Engl J Med* 2013;368(19):1771–80.
- [89] Wandt H, Schaefer-Eckart K, Wendelin K, et al. Therapeutic platelet transfusion versus routine prophylactic transfusion in patients with hematological malignancies: an open-label, multicentre, randomised study. *Lancet Lond Engl* 2012;380(9850):1309–16.
- [90] Estcourt LJ, Stanworth S, Doree C, et al. Different doses of prophylactic platelet transfusion for preventing bleeding in patients with haematological disorders after chemotherapy or stem cell transplantation. *Cochrane Database Syst Rev* 2014;(3):CD010984.
- [91] Estcourt LJ, Birchall J, Allard S, et al. Guidelines for the use of platelet transfusions. *Br J Haematol* 2017;176(3):365–94.
- [92] Darmon M, Bourmaud A, Georges Q, et al. Changes in critically ill cancer patients short-term outcome over the last decades: results of systematic review with meta-analysis on individual data. *Intensive Care Med* 2019;45:977–87.
- [93] Azoulay E, Pène F, Darmon M, et al. Managing critically ill hematology patients: Time to think differently. *Blood Rev* 2015;29(6):359–67.
- [94] Assoun S, Lemiale V, Azoulay E. Molecular targeted therapy-related life-threatening toxicity in patients with malignancies. A systematic review of published cases. *Intensive Care Med* 2019;45(7):988–97.
- [95] Lee CJ, Wang T, Chen K, et al. Severity of chronic graft-versus-host disease and late effects following allogeneic hematopoietic cell transplantation for adults with hematologic malignancy. *Transplant Cell Ther* 2024;30(1):97.e1–97.e14.
- [96] Lecuyer L, Chevret S, Guidet B, et al. Case volume and mortality in haematological patients with acute respiratory failure. *Eur Respir J* 2008;32:748–54.
- [97] Azoulay E, Pickkers P, Soares M, et al. Acute hypoxemic respiratory failure in immunocompromised patients: the Efrain multinational prospective cohort study. *Intensive Care Med* 2017. *Intensive Care Med*. 2017 Dec;43(12):1808–19. <https://doi.org/10.1007/s00134-017-4947-1>. Epub 2017 Sep 25.
- [98] LoCastro M, Wang Y, Yu T, et al. Clinicians' perspectives on the telehealth serious illness care program for older adults with myeloid malignancies: single-arm pilot study. *JMIR Form Res* 2024;8:e58503.
- [99] Vincent J-L. Withdrawing may be preferable to withholding. *Crit Care Lond Engl* 2005;9(3):226–9.
- [100] Vink EE, Azoulay E, Caplan A, Kompanje EJO, Bakker J. Time-limited trial of intensive care treatment: an overview of current literature. *Intensive Care Med* 2018;44(9):1369–77.
- [101] Kruser JM, Nadig NR, Vigilanti EM, et al. Time-limited trials for patients with critical illness: a review of the literature. *Chest* 2024;165(4):881–91.
- [102] Kruser JM, Ashana DC, Courtright KR, et al. Defining the time-limited trial for patients with critical illness: an official american thoracic society workshop report. *Ann Am Thorac Soc* 2024;21(2):187–99.
- [103] Shrimel MG, Ferket BS, Scott DJ, et al. Time-limited trials of intensive care for critically ill patients with cancer: how long is long enough? *JAMA Oncol* 2016;2(1):76–83.
- [104] Berntzen H, Rustøen T, Kynø NM. “Hope at a crossroads” - Experiences of hope in intensive care patients: a qualitative study. *Aust Crit Care Off J Confed Aust Crit Care Nurses* 2024;37(1):120–6.
- [105] Smith AK, White DB, Arnold RM. Uncertainty—the other side of prognosis. *N Engl J Med* 2013;368(26):2448–50.
- [106] Nunez ER, Schenker Y, Joel ID, et al. Acutely bereaved surrogates' stories about the decision to limit life support in the ICU. *Crit Care Med* 2015;43(11):2387–93.
- [107] Patel B, Gheihman G, Katz JT, Begin AS, Solomon SR. Navigating uncertainty in clinical practice: a structured approach. *J Gen Intern Med* 2024;39(5):829–36.
- [108] Epstein AS, Desai AV, Bernal C, et al. Giving voice to patient values throughout cancer: a novel nurse-led intervention. *J Pain Symptom Manag* 2019;58(1):72–79.e2.
- [109] Gilligan T, Coyle N, Frankel RM, et al. Patient-clinician communication: american society of clinical oncology consensus guideline. *J Clin Oncol Off J Am Soc Clin Oncol* 2017;35(31):3618–32.
- [110] Rosa WE, Epstein AS, Nelson JE. A values proposition for cancer care. *JAMA Oncol* 2024. *JAMA Oncol* 2025 Feb 1;11(2):107–8. <https://doi.org/10.1001/jamaoncol.2024.5572>.
- [111] Chochinov HM. The platinum rule: a new standard for person-centered care. *J Palliat Med* 2022;25(6):854–6.
- [112] Chochinov HM, McClement S, Hack T, et al. Eliciting personhood within clinical practice: effects on patients, families, and health care providers. *J Pain Symptom Manag* 2015;49(6):974–980.e2.
- [113] Feudtner C. The breadth of hopes. *N Engl J Med* 2009;361(24):2306–7.
- [114] van Vliet M, van den Boogaard M, Donnelly JP, et al. Long-term health related quality of life following intensive care during treatment for haematological malignancies. *PLoS ONE* 2014;9(1):e87779.
- [115] Nelson JE, Meier DE, Oei EJ, et al. Self-reported symptom experience of critically ill cancer patients receiving intensive care. *Crit Care Med* 2001;29(2):277–82.

- [116] Puntillo KA, Arai S, Cohen NH, et al. Symptoms experienced by intensive care unit patients at high risk of dying. *Crit Care Med* 2010;38(11):2155–60.
- [117] Saltnes-Lillegård C, Rustøen T, Beitland S, et al. Self-reported symptoms experienced by intensive care unit patients: a prospective observational multicenter study. *Intensive Care Med* 2023;49(11):1370–82.
- [118] Demoule A, Decavele M, Antonelli M, et al. Dyspnoea in acutely ill mechanically ventilated adult patients: an ERS/ESICM statement. *Intensive Care Med* 2024;50(2):159–80.
- [119] Kalfon P, Alessandrini M, Boucekine M, et al. Tailored multicomponent program for discomfort reduction in critically ill patients may decrease post-traumatic stress disorder in general ICU survivors at 1 year. *Intensive Care Med* 2019;45: 223–35.
- [120] Puntillo K, Nelson JE, Weissman D, et al. Palliative care in the ICU: relief of pain, dyspnea, and thirst—a report from the IPAL-ICU Advisory Board. *Intensive Care Med* 2014;40(2):235–48.
- [121] Azoulay E, Russell L, Van de Louw A, et al. Diagnosis of severe respiratory infections in immunocompromised patients. *Intensive Care Med* 2020;46(2): 298–314.
- [122] Legrand M, Max A, Peigne V, et al. Survival in neutropenic patients with severe sepsis or septic shock. *Crit Care Med* 2012;40(1):43–9.
- [123] Pardo E, Lemiale V, Mokart D, et al. Invasive pulmonary aspergillosis in critically ill patients with hematological malignancies. *Intensive Care Med* 2019;45(12): 1732–41.
- [124] Jestin M, Azoulay E, Pène F, et al. Poor outcome associated with mucormycosis in critically ill hematological patients: results of a multicenter study. *Ann Intensive Care* 2021;11(1):31.
- [125] Garcia Garrido HM, Knol MJ, Heijmans J, et al. Invasive pneumococcal disease among adults with hematological and solid organ malignancies: a population-based cohort study. *Int J Infect Dis IJID Off Publ Int Soc Infect Dis* 2021;106: 237–45.
- [126] Fernández S, Castro P, Azoulay E. What intensivists need to know about cytomegalovirus infection in immunocompromised ICU patients. *Intensive Care Med* 2025. *Intensive Care Med* 2025 Jan;51(1):39–61. <https://doi.org/10.1007/s00134-024-07737-5>. Epub 2025 Jan 7.
- [127] Bennett C, Vardiman J, Golomb H. Disseminated atypical mycobacterial infection in patients with hairy cell leukemia. *Am J Med* 1986;80(5):891–6.
- [128] Maffei R, Maccaferri M, Arletti L, et al. Immunomodulatory effect of ibrutinib: reducing the barrier against fungal infections. *Blood Rev* 2020;40:100635.
- [129] Schnell D, Azoulay E, Benoit D, et al. Management of neutropenic patients in the intensive care unit (NEWBORNS EXCLUDED) recommendations from an expert panel from the French Intensive Care Society (SRLF) with the French Group for Pediatric Intensive Care Emergencies (GFRUP), the French Society of Anesthesia and Intensive Care (SFAR), the French Society of Hematology (SFH), the French Society for Hospital Hygiene (SF2H), and the French Infectious Diseases Society (SPILF). *Ann Intensive Care* 2016;6:90.
- [130] Hill JA, Li D, Hay KA, et al. Infectious complications of CD19-targeted chimeric antigen receptor-modified T-cell immunotherapy. *Blood* 2018;131(1):121–30.
- [131] Mariotte E, Azoulay E, Galicier L, et al. Epidemiology and pathophysiology of adulthood-onset thrombotic microangiopathy with severe ADAMTS13 deficiency (thrombotic thrombocytopenic purpura): a cross-sectional analysis of the French national registry for thrombotic microangiopathy. *Lancet Haematol* 2016;3(5): e237–45.
- [132] Freifeld AG, Bow EJ, Sepkowitz KA, et al. Clinical practice guideline for the use of antimicrobial agents in neutropenic patients with cancer: 2010 update by the infectious diseases society of america. *Clin Infect Dis Off Publ Infect Dis Soc Am* 2011;52(4):e56–93.
- [133] Lopez R, Rello J, Taccone F, et al. Aminoglycosides in immunocompromised critically ill patients with bacterial pneumonia and septic shock: a post-hoc analysis of a prospective multicenter multinational cohort. *Shock* 2020;54:731–7.
- [134] Cometta A, Kern WV, De Bock R, et al. Vancomycin versus placebo for treating persistent fever in patients with neutropenic cancer receiving piperacillin-tazobactam monotherapy. *Clin Infect Dis Off Publ Infect Dis Soc Am* 2003;37(3): 382–9.
- [135] Schnell D, Montlahuc C, Bruneel F, et al. De-escalation of antimicrobial therapy in critically ill hematology patients: a prospective cohort study. *Intensive Care Med* 2019;45:743–5.
- [136] Averbuch D, Orasch C, Cordonnier C, et al. European guidelines for empirical antibacterial therapy for febrile neutropenic patients in the era of growing resistance: summary of the 2011 4th European Conference on Infections in Leukemia. *Haematologica* 2013;98(12):1826–35.
- [137] Stern A, Carrara E, Bitterman R, et al. Early discontinuation of antibiotics for febrile neutropenia versus continuation until neutropenia resolution in people with cancer. *Cochrane Database Syst Rev* 2019;1(1):CD012184.
- [138] Denning DW. Global incidence and mortality of severe fungal disease. *Lancet Infect Dis* 2024. S1473–3099(23), 00692–8. *Lancet Infect Dis* 2024 Jul;24(7): e428–38.
- [139] Mellinghoff SC, Panse J, Alakel N, et al. Primary prophylaxis of invasive fungal infections in patients with hematological malignancies: 2017 update of the recommendations of the Infectious Diseases Working Party (AGIHO) of the German Society for Haematology and Medical Oncology (DGHO). *Ann Hematol* 2018;97(2):197–207.
- [140] Lortholary O, Renaudat C, Sitbon K, et al. Worrisome trends in incidence and mortality of candidemia in intensive care units (Paris area, 2002–2010). *Intensive Care Med* 2014;40(9):1303–12.
- [141] Donnelly J, Chen S, Kauffman C, et al. Revision and update of the consensus definitions of invasive fungal disease from the European organization for research and treatment of cancer and the mycoses study group education and research consortium. *Clin Infect Dis* 2020;71:1367–76.
- [142] Rocchi S, Scherer E, Mengoli C, et al. Interlaboratory evaluation of Mucorales PCR assays for testing serum specimens: a study by the fungal PCR Initiative and the Modimucor study group. *Med Mycol* 2021;59(2):126–38.
- [143] Cornely OA, Arikian-Akdagli S, Dannaoui E, et al. ESCMID and ECMM joint clinical guidelines for the diagnosis and management of mucormycosis 2013. *Clin Microbiol Infect Off Publ Eur Soc Clin Microbiol Infect Dis* 2014;20(Suppl. 3): 5–26.
- [144] Cornely OA, Böhme A, Buchheidt D, et al. Prophylaxis of invasive fungal infections in patients with hematological malignancies and solid tumors—guidelines of the Infectious Diseases Working Party (AGIHO) of the German Society of Hematology and Oncology (DGHO). *Ann Hematol* 2003;82(Suppl. 2): S186–200.
- [145] Mokart D, Slehofer G, Lambert J, et al. De-escalation of antimicrobial treatment in neutropenic patients with severe sepsis: results from an observational study. *Intensive Care Med* 2014;40(1):41–9.
- [146] Kriegl L, Egger M, Boyer J, Hoenigl M, Krause R. New treatment options for critically important WHO fungal priority pathogens. *Clin Microbiol Infect Off Publ Eur Soc Clin Microbiol Infect Dis* 2024;S1198-743X(24)00118–6.
- [147] Burghi G, Biard L, Roux A, et al. Characteristics and outcome according to underlying disease in non-AIDS patients with acute respiratory failure due to pneumocystis pneumonia. *Eur J Clin Microbiol Infect Dis* 2021;40:1191–8.
- [148] Giacobbe DR, Dettori S, Di Pilato V, et al. Pneumocystis jirovecii pneumonia in intensive care units: a multicenter study by ESGCIP and EFISG. *Crit Care Lond Engl* 2023;27(1):323.
- [149] Alanio A, Hauser PM, Lagrou K, et al. ECIL guidelines for the diagnosis of Pneumocystis jirovecii pneumonia in patients with hematological malignancies and stem cell transplant recipients. *J Antimicrob Chemother* 2016;71(9): 2386–96.
- [150] Brusosa M, Ruiz S, Monge I, et al. Impact of letermovir prophylaxis in CMV reactivation and disease after allogeneic hematopoietic cell transplantation: a real-world, observational study. *Ann Hematol* 2024;103(2):609–21.
- [151] Baber A, Calvet L, Vissac C, et al. Cytomegalovirus infection in intensive care unit patients with hematological malignancies: characteristics and clinical outcomes. *J Crit Care* 2024;82:154766.
- [152] Gonzalez F, Beschmout S, Chow-Chine L, et al. Herpesviridae in critically ill hematology patients: HHV-6 is associated with worse clinical outcome. *J Crit Care* 2021;66:138–45.
- [153] Fernández S, Grafia I, Peyrony O, et al. Clinical characteristics and outcomes of immunocompromised critically ill patients with cytomegalovirus end-organ disease: a multicenter retrospective cohort study. *Crit Care Lond Engl* 2024;28(1): 243.
- [154] Ward KN, Hill JA, Hubacek P, et al. Guidelines from the 2017 European Conference on Infections in Leukaemia for management of HHV-6 infection in patients with hematologic malignancies and after hematopoietic stem cell transplantation. *Haematologica* 2019;104(11):2155–63.
- [155] Matthes-Martin S, Feuchtinger T, Shaw PJ, et al. European guidelines for diagnosis and treatment of adenovirus infection in leukemia and stem cell transplantation: summary of ECIL-4 (2011). *Transpl Infect Dis Off J Transplant Soc* 2012;14(6):555–63.
- [156] Kochanek M, Schalk E, von Bergwelt-Baildon M, et al. Management of sepsis in neutropenic cancer patients: 2018 guidelines from the Infectious Diseases Working Party (AGIHO) and Intensive Care Working Party (ICHOP) of the German Society of Hematology and Medical Oncology (DGHO). *Ann Hematol* 2019;98(5):1051–69.
- [157] Doukhan L, Bisbal M, Chow-Chine L, et al. Respiratory events in ward are associated with later intensive care unit (ICU) admission and hospital mortality in onco-hematology patients not admitted to ICU after a first request. *PLoS ONE* 2017;12(7):e0181808.
- [158] Rivera DR, Gallicchio L, Brown J, et al. Trends in adult cancer-related emergency department utilization: an analysis of data from the nationwide emergency department sample. *JAMA Oncol* 2017;3(10):e172450.